Task result overview

Task information

Software version: 1.2.3

Generated UTC: 2026-10-10T02:28:34Z

Task name: reference_600km_24h_report

Analysis duration: 24 h

Time step: 60 s

Thermal network: reference_11_node_gnss_receiver

Outputs from this run

scenario

Main findings

Mean beta angle: 19.46 deg

Eclipse fraction: 36.29 %

RF temperature span: 5.183 K

Semi-synthetic code-bias span: 1.12 m

Heat-to-RF response lag: Not available s

Maximum energy closure: 1.775e-09 J

Interpretation

Temperature and lag describe this scenario; interpret them with the inputs and plots.

Code bias comes from assumed thermal sensitivity, not measured GNSS error.

Orbit heating and nodal response

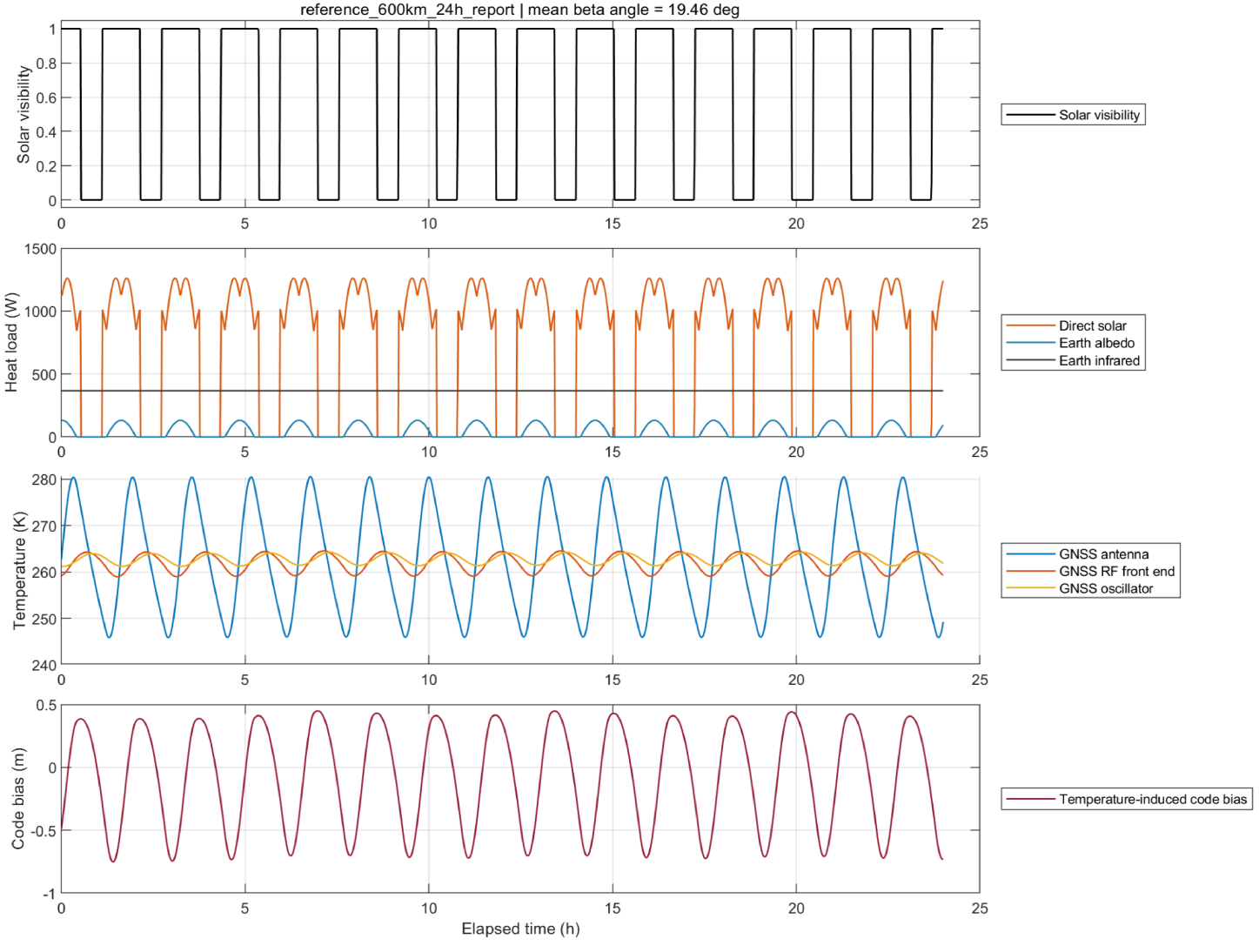
Mean beta angle: 19.46 deg; eclipse fraction: 36.29%.

RF temperature span: 5.183 K; assumed temperature-induced code-bias span: 1.12 m.

The lag metric is undefined in this run, not zero lag.

30 illumination transitions occurred over this run.

The code bias uses assumed thermal sensitivity; it is not measured GNSS error.



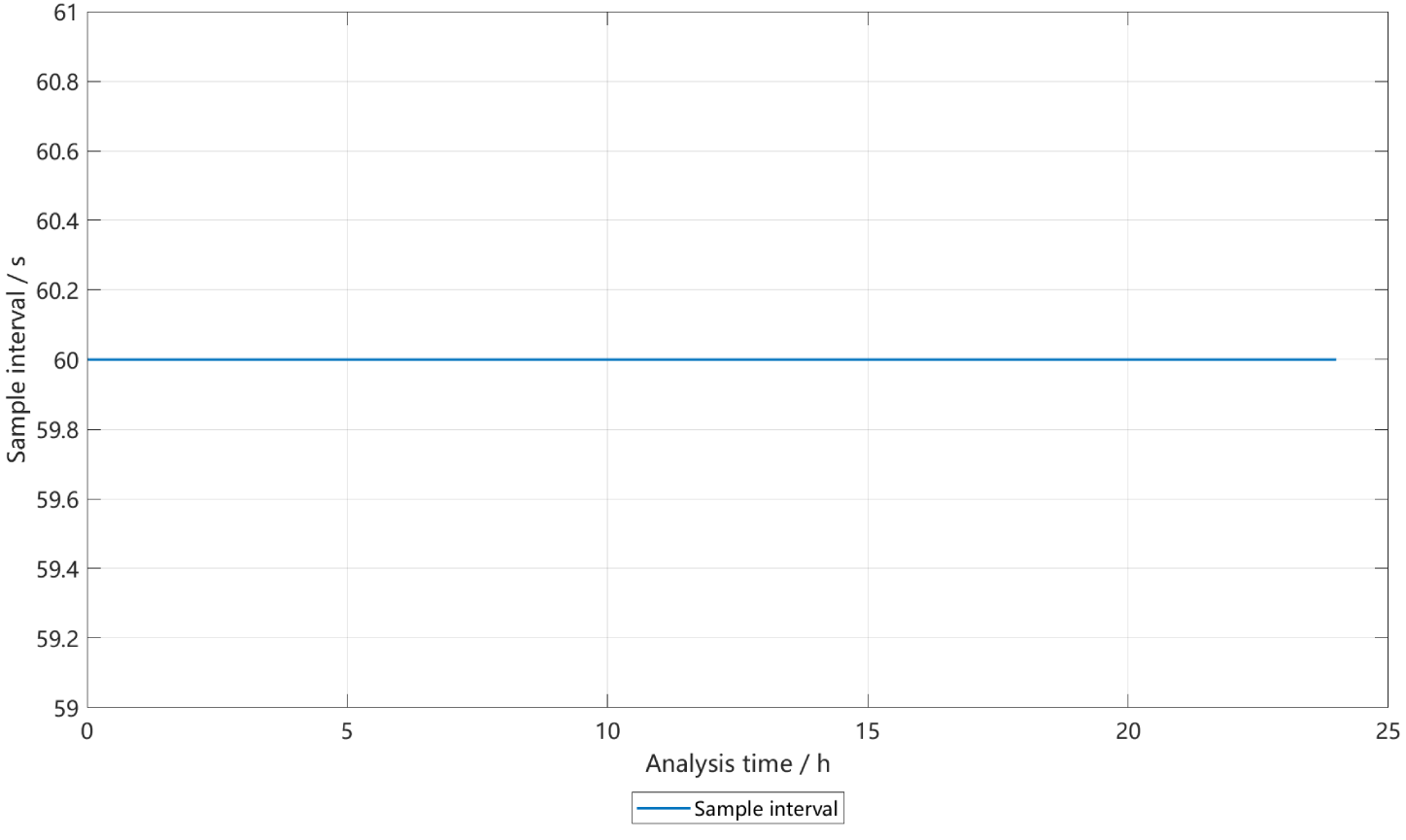
Time coverage and resolution

1441 retained epochs span 24.000 h, approximately 14.89 orbits.

Output spacing: 60 to 60 s. No missing observations are filled.

The forcing grid limits eclipse-edge and peak timing resolution, even with smaller solver steps.

Less than one orbit or unidentifiable peaks cannot establish periodic equilibrium or a reliable lag.



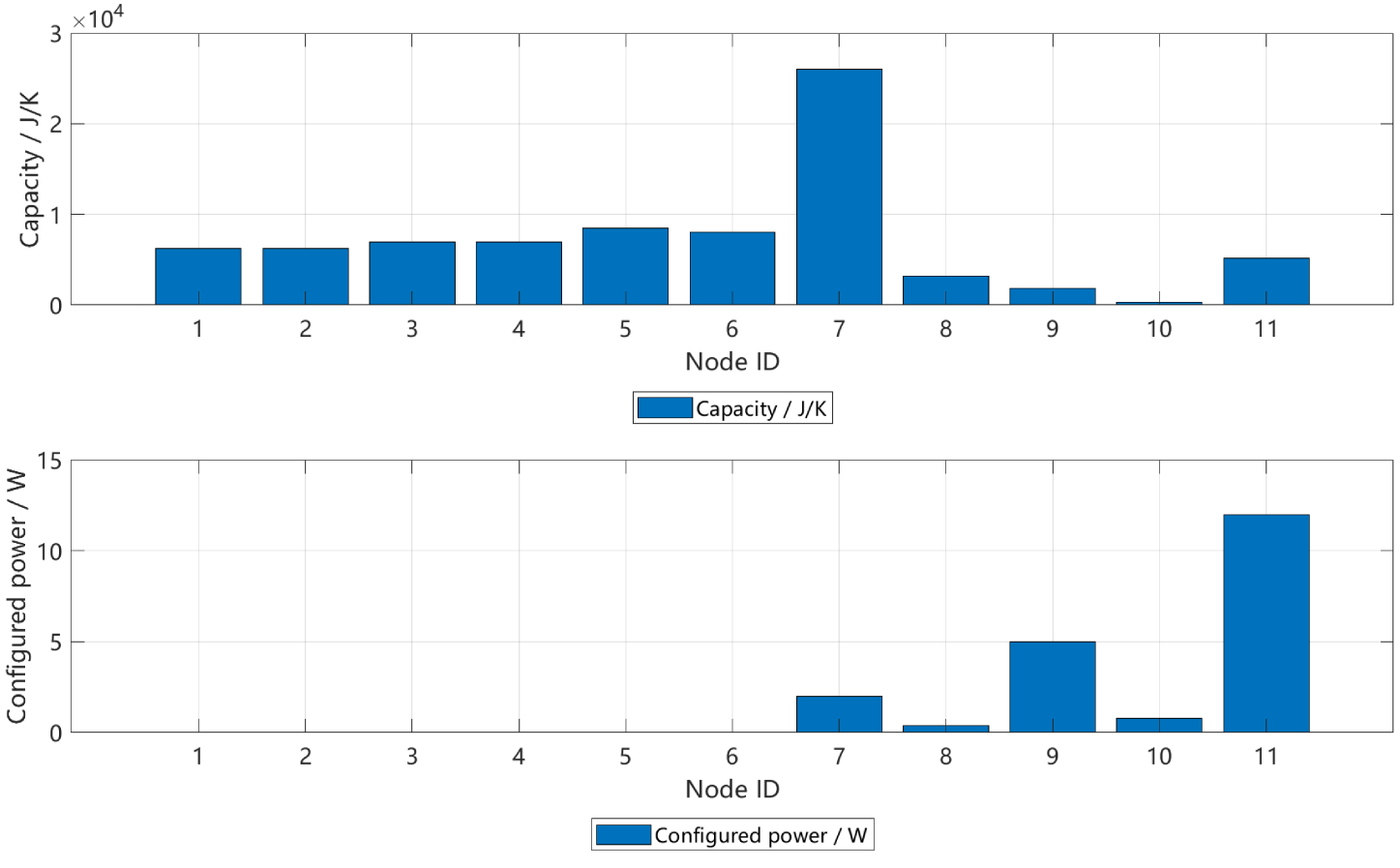
Thermal capacity and internal power

11 nodes; total capacity 79420 J/K; configured internal power 20.2 W.

Capacity controls the rate of temperature change for a given net heat input; power affects the baseline.

These are configured values. Neither capacity nor power alone determines a thermal time constant.

Plots select 11/11 nodes by full temperature range; IDs and all nodes are tabulated.



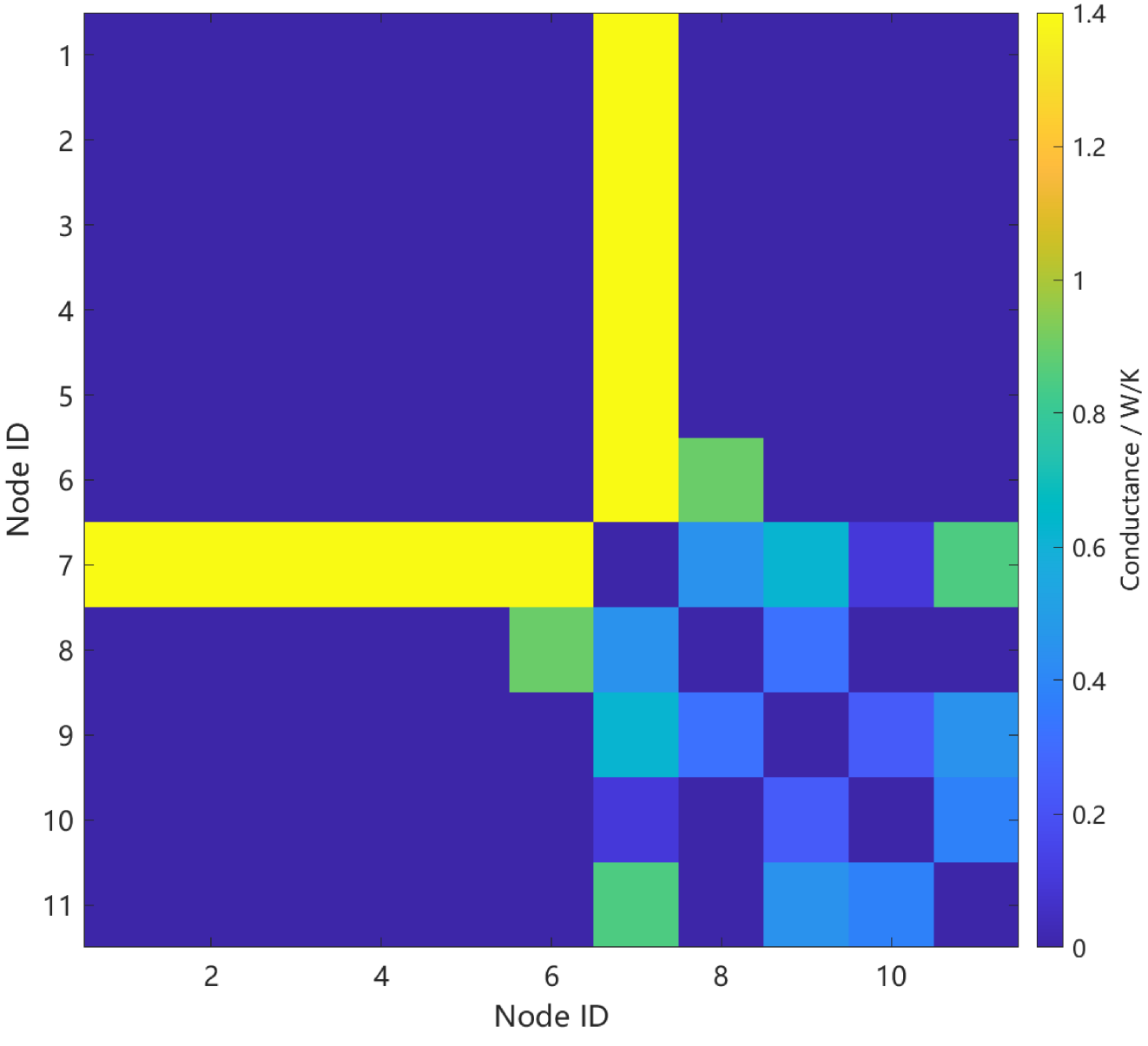
Thermal links and conductance

15 nonzero undirected links; the first 11 nodes are shown.

Nonzero entries are link conductances in W/K, not material conductivity in W/(m K).

Instantaneous link heat flow is conductance times temperature difference.

The configured links require device geometry, material and contact evidence for physical validation.



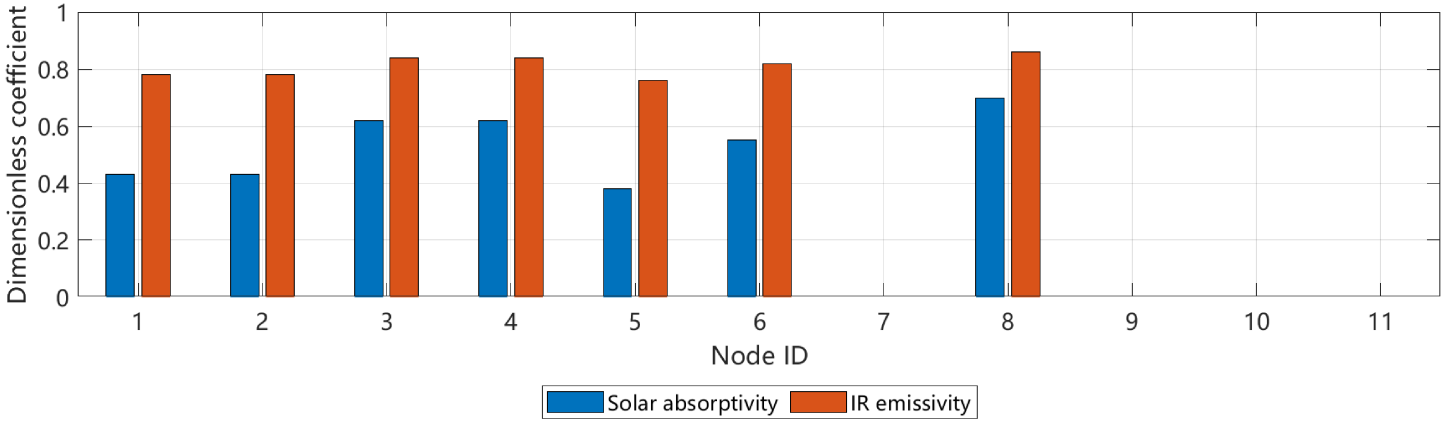
Optical properties and radiating area

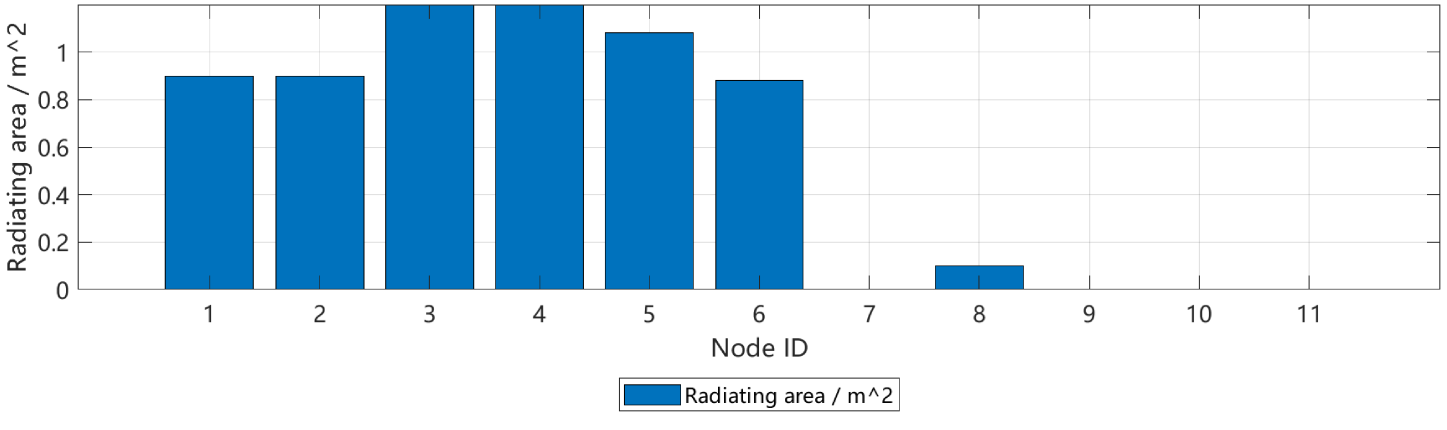
Solar absorptivity controls shortwave absorption; emissivity and area control radiation.

Absorptivity and emissivity refer to different spectral bands and are not interchangeable.

These are equivalent nodal properties, not proof of facet shadowing or material-test validation.

Plots select 11/11 nodes by full temperature range; IDs and all nodes are tabulated.





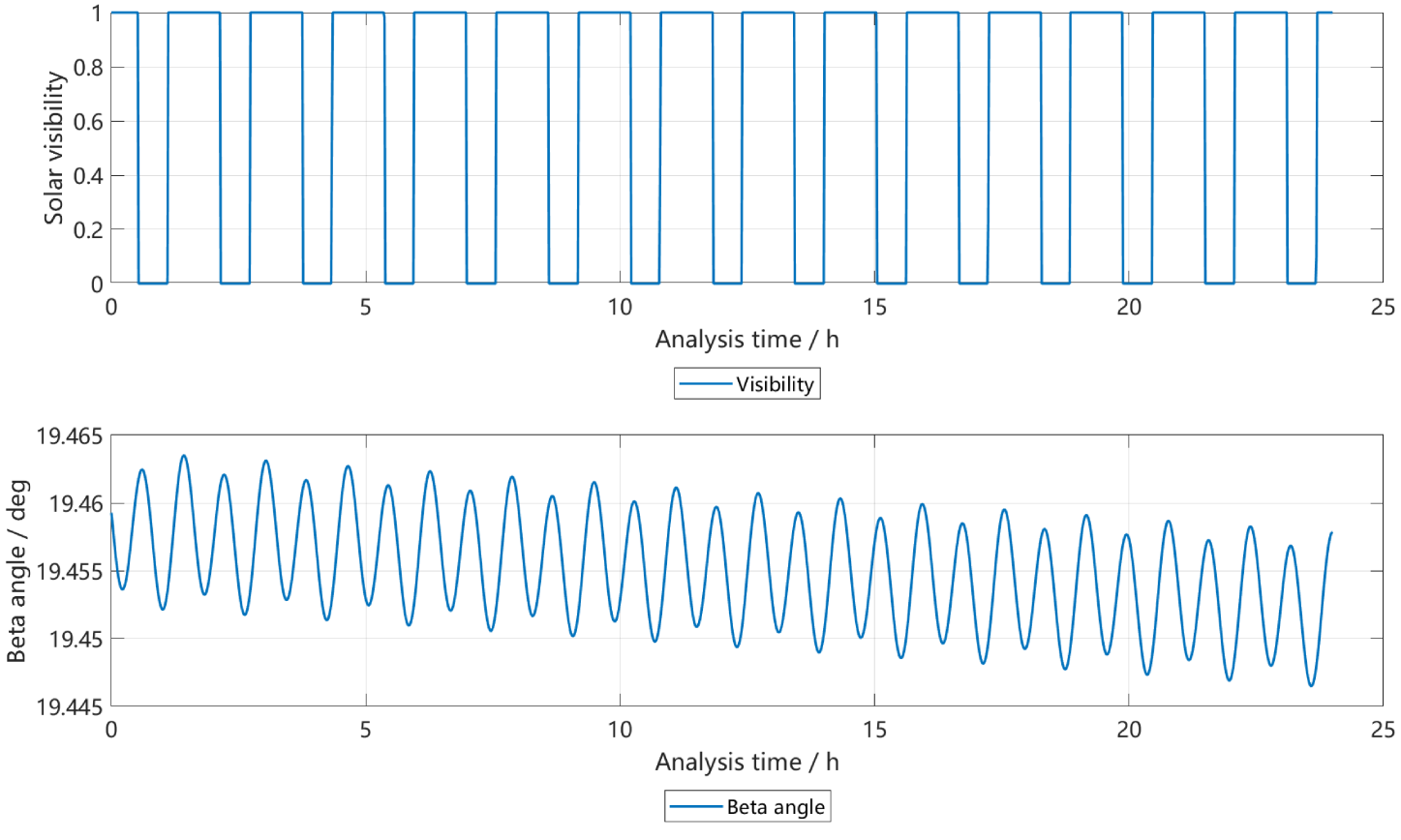
Illumination and eclipse structure

Trapezoidal time weighting estimates partial/full eclipse over 36.32% of the window.

Solar visibility ranges from 0.000 to 1.000; orbital period 96.69 min.

Eclipse uses visibility below 0.999. Time-weighted and epoch-count fractions can differ.

No eclipse does not imply constant temperature: attitude changes can alter solar incidence.



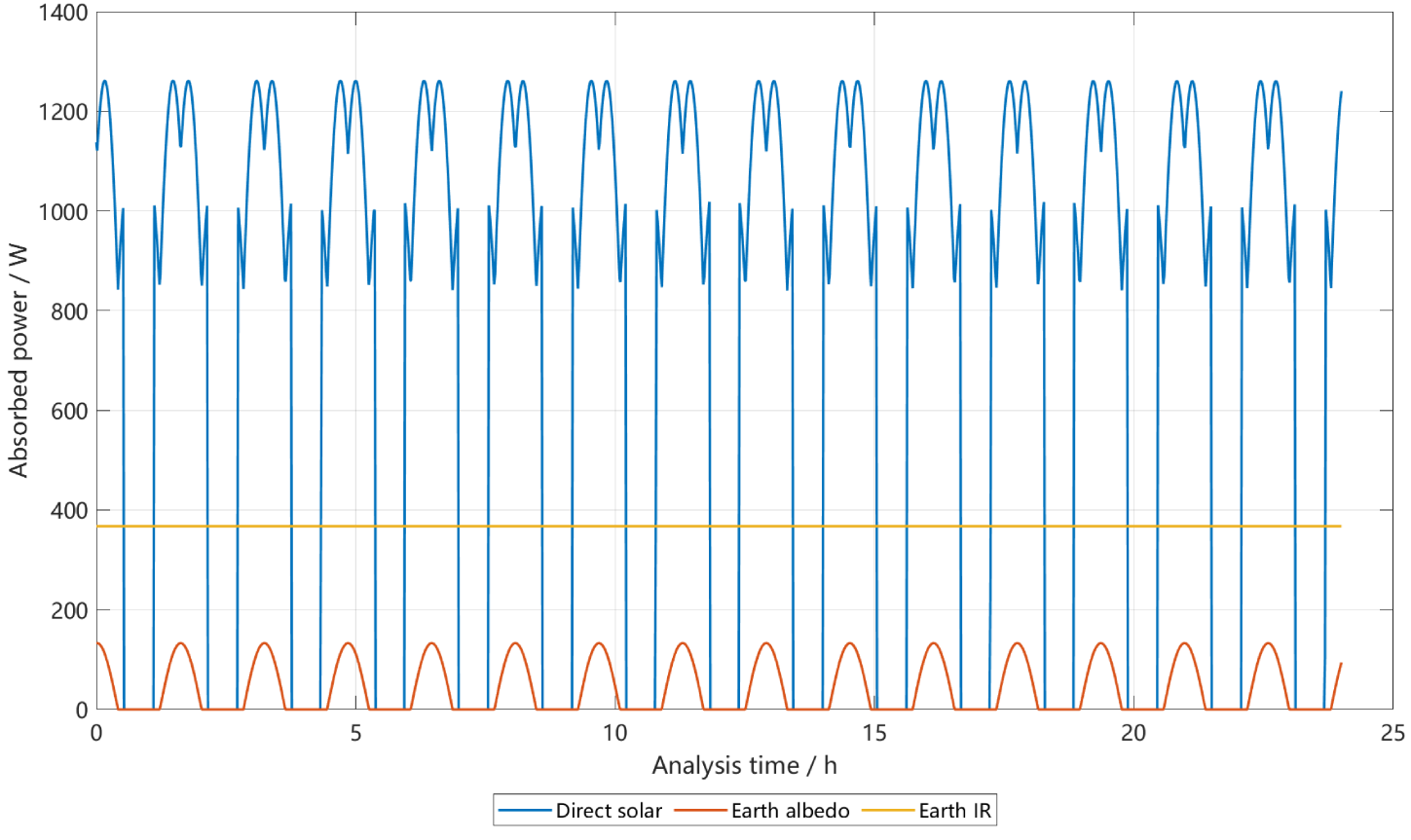
Environmental heat inputs

Time-mean absorbed power: solar 704.069 W; albedo 41.743 W; Earth IR 367.422 W.

Curves sum absorbed power over the network, including area and absorption properties.

Input power is not temperature; response also depends on capacity, conduction and radiation.

Direct sunlight decreases in eclipse; Earth infrared does not automatically vanish.



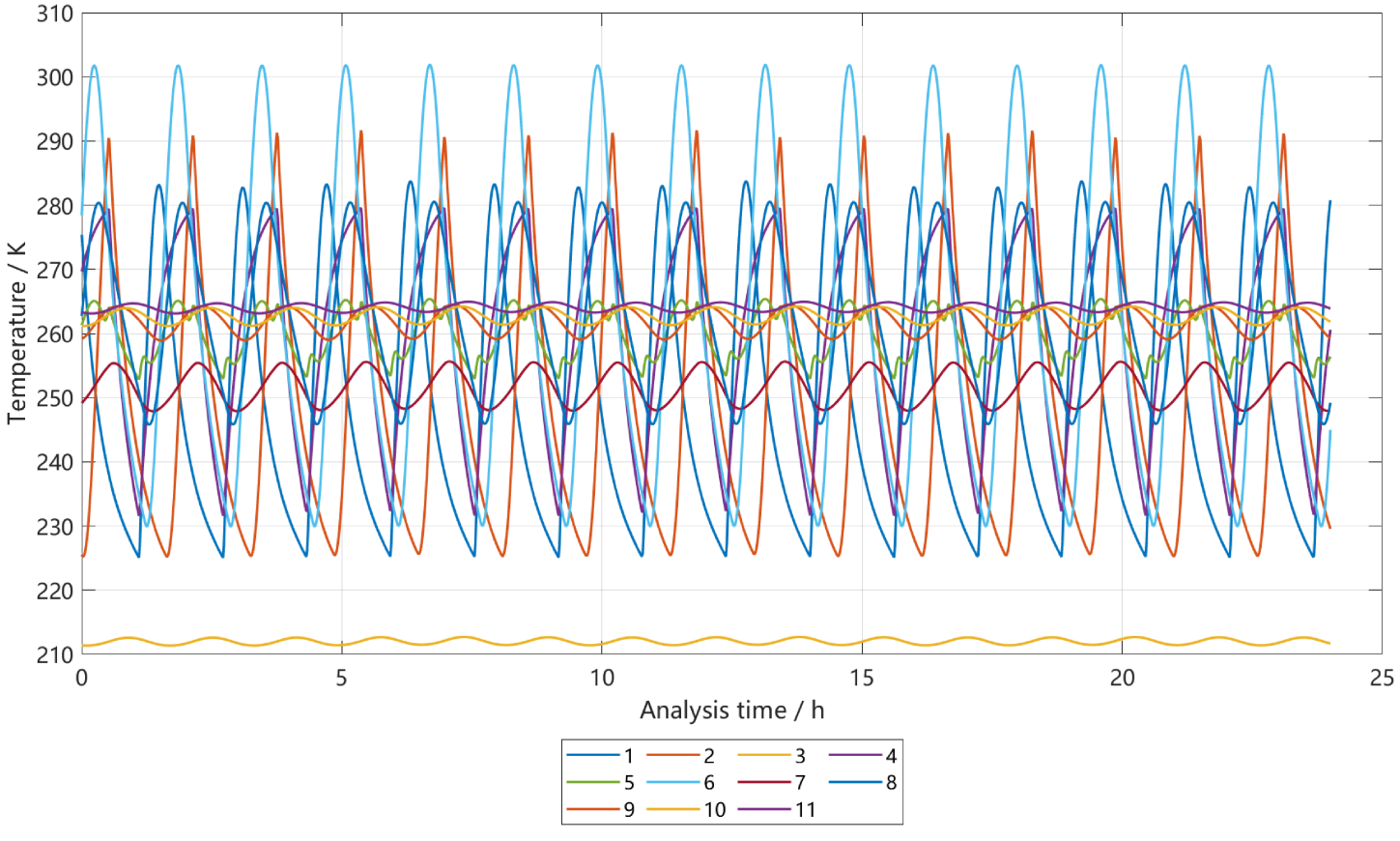
Nodal temperatures and extrema

Window maximum 301.864 K (28.71 deg C, node 6); minimum 211.359 K (-61.79 deg C, node 3).

Largest full peak-to-peak range 71.952 K (node 6), distinct from a 95th-5th percentile span.

Without device operating limits, extrema cannot establish engineering compliance.

Plots select 11/11 nodes by full temperature range; IDs and all nodes are tabulated.



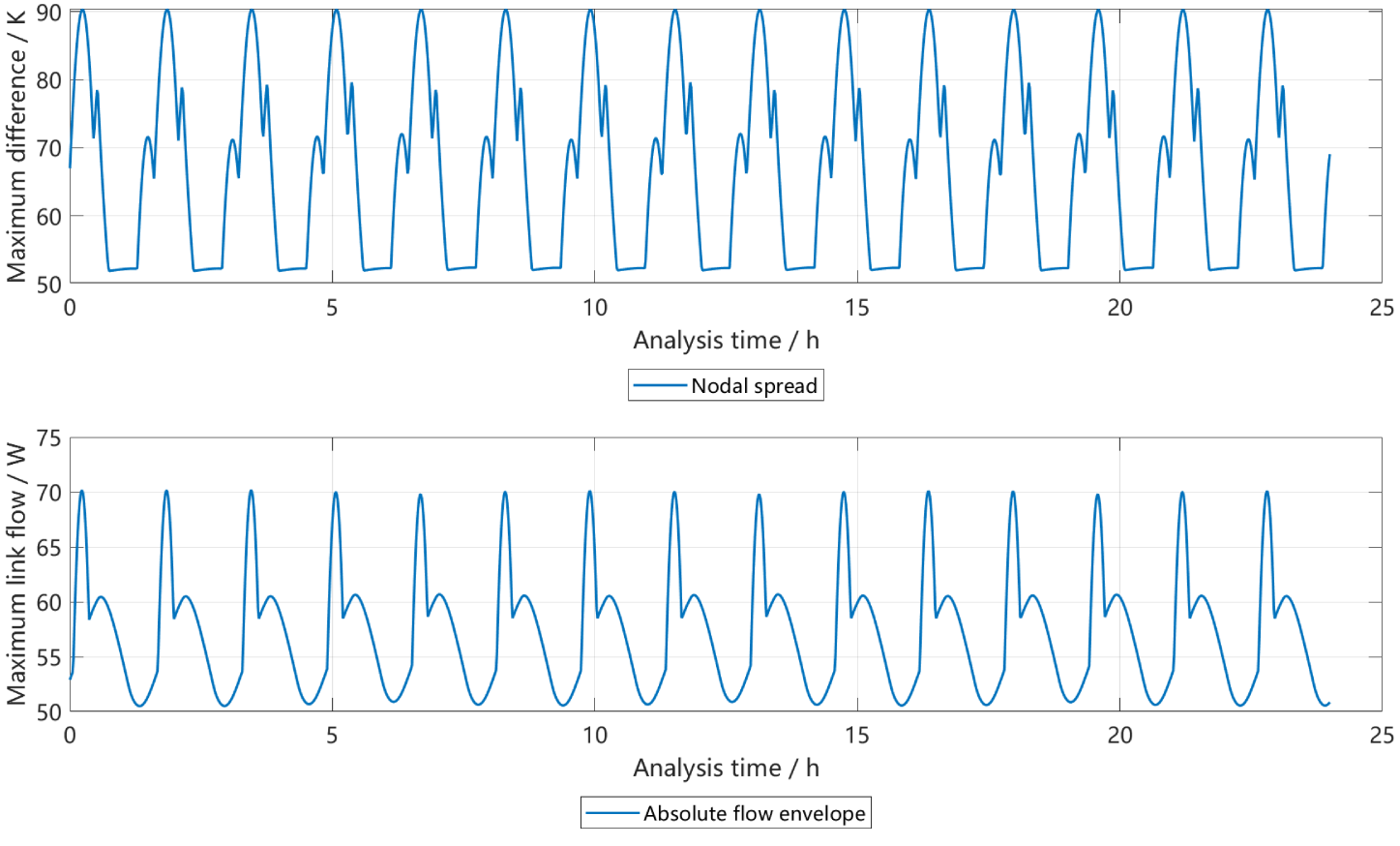
Temperature differences and link heat flow

Maximum simultaneous nodal spread 90.341 K; maximum absolute single-link flow 70.175 W.

Panels show epoch-wise temperature spread and the largest absolute link heat flow.

The dominant link may change; internal link transfers are not external heat inputs.

Nodal differences are not 3D temperature gradients or direct evidence of stress safety.



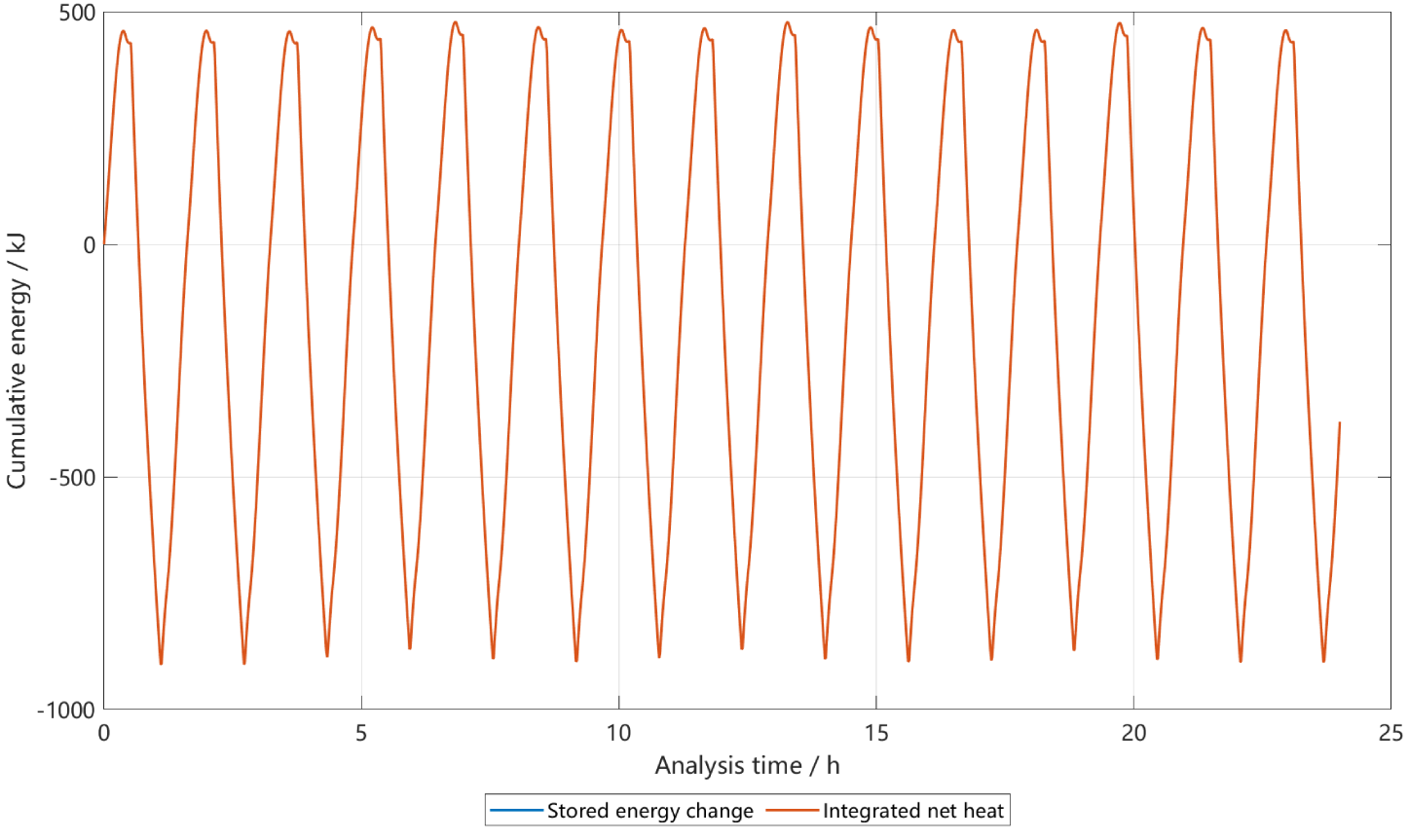
Stored energy and energy balance

After rebasing both at the analysis start, maximum cumulative closure is 4.8938e-08 J.

Stored energy is the sum of capacity times temperature change, relative to analysis start.

Balance: stored energy = external heat + internal heat - radiation + boundary heat.

Internal conduction cancels in the network sum. Closure is not grid convergence or physical validation.



Thermal lag and identifiability

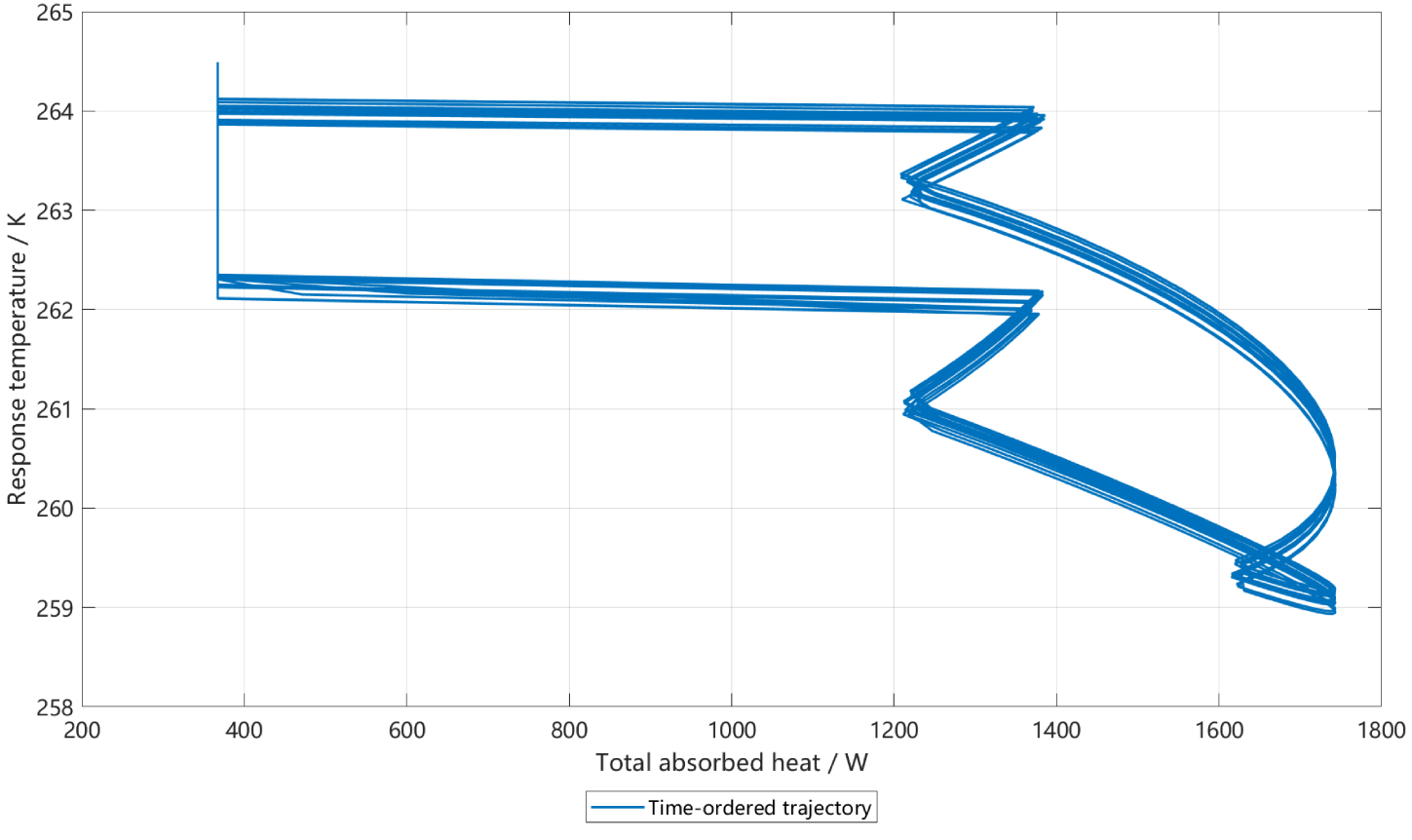
Response node 9; scored cycles 14; nonnegative candidates 11; mixed phase signs 1.

Peak lag NaN s; consistent-sign median phase NaN s; sampling resolution 60 s.

Mixed wrapped-phase signs invalidate the aggregate lag; positive cycles are not selectively pooled.

Undefined is not zero lag. A peak-to-peak delay is not a thermal time constant.

The sampled forcing-temperature trajectory may include initial transients; its shape alone is not mechanism proof.



Initial state and temperature drift

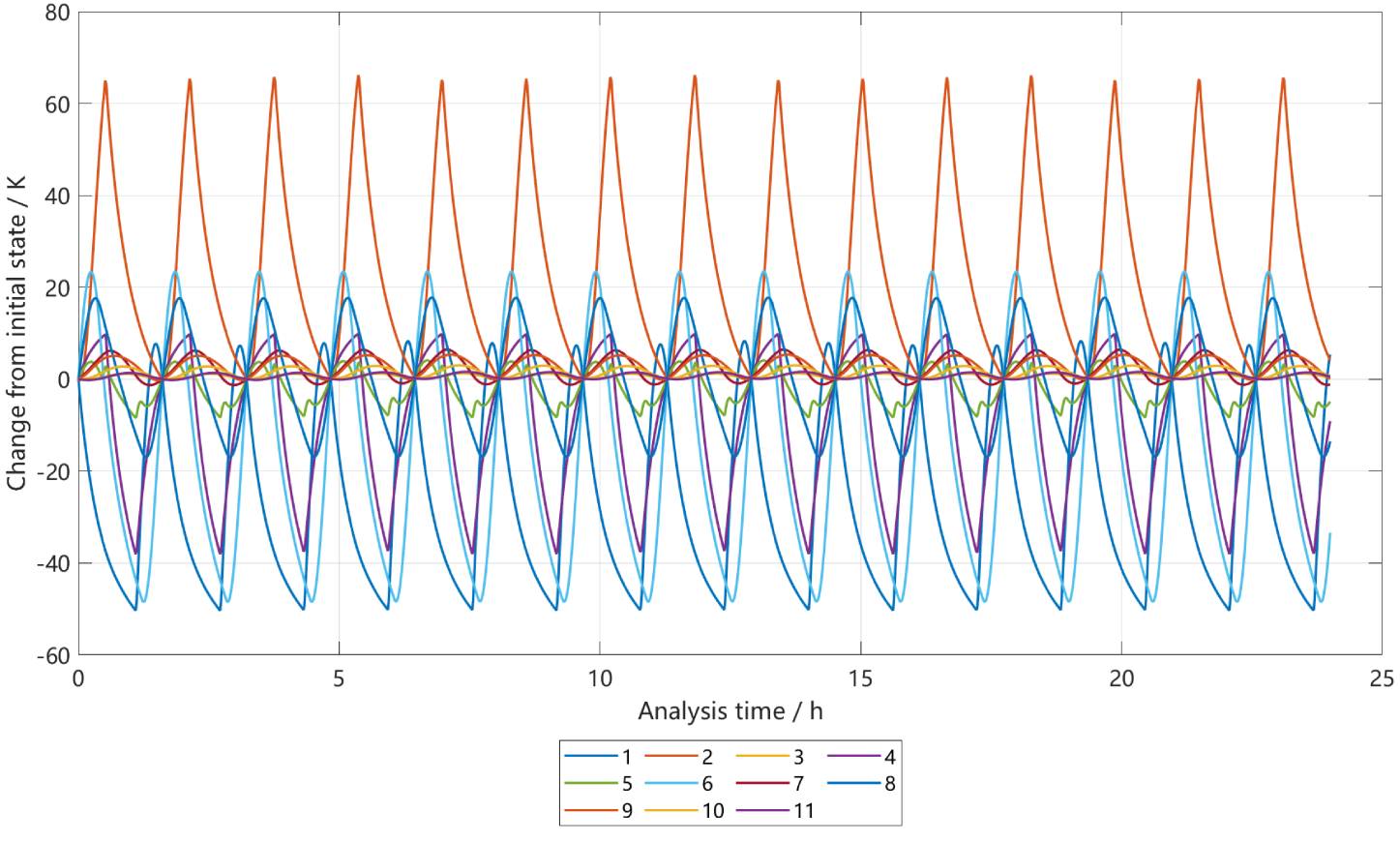
Frozen-Sun initialization: 17 cycles; convergence flag 1; error 0.00045696 K.

Largest absolute end-minus-start temperature change 33.415 K.

Endpoints may have different orbital phases, so their difference is not periodic-state error.

Equilibrium needs same-phase cycle comparison; accuracy needs a separate grid-refinement test.

Plots select 11/11 nodes by full temperature range; IDs and all nodes are tabulated.



Time-grid refinement audit

Independent reruns keep the model and scenario fixed and compare exact common epochs.

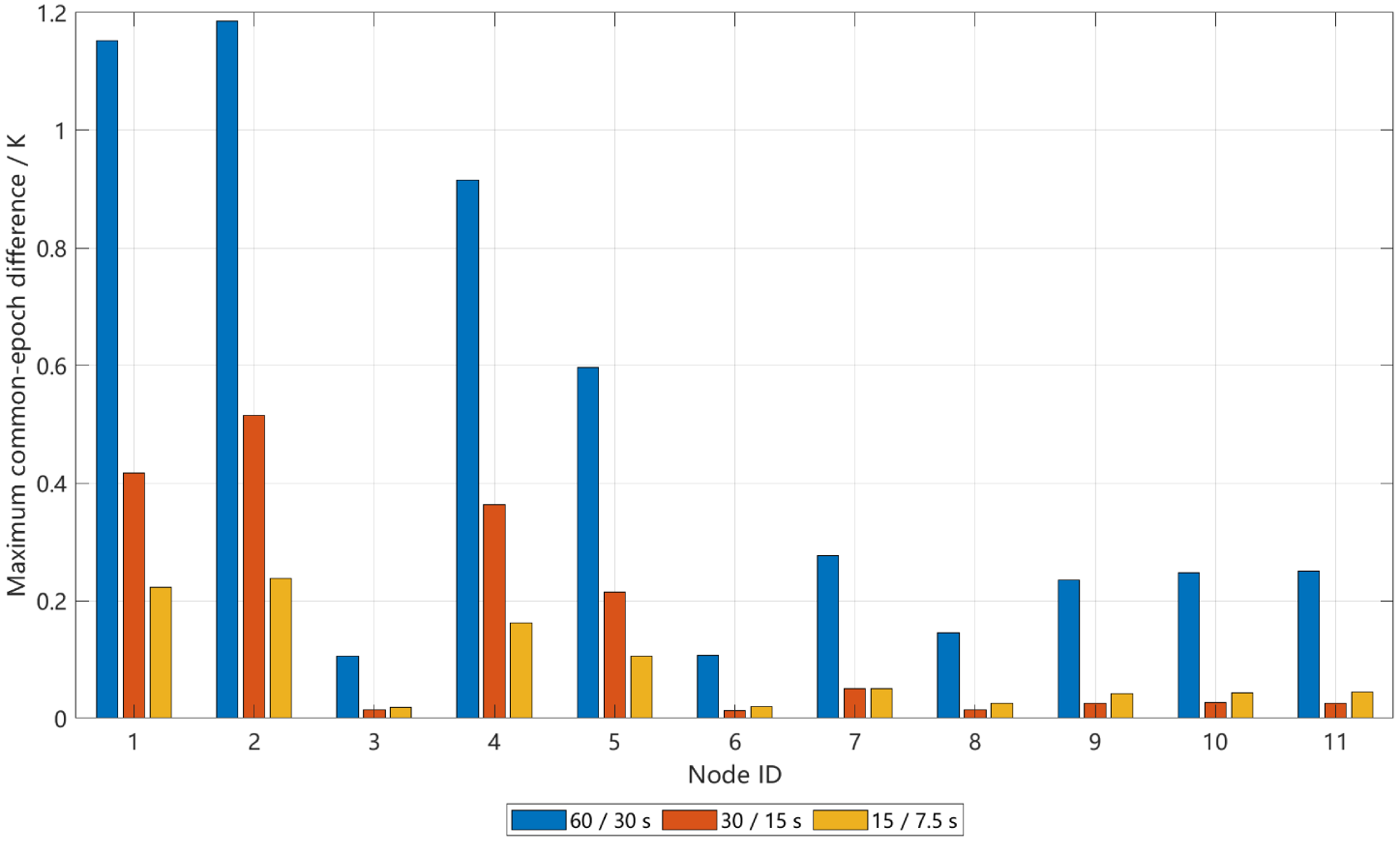
60/30 s grids: maximum difference 1.186 K; response-node difference 0.2349 K.

30/15 s grids: maximum difference 0.5155 K; response-node difference 0.0256 K.

15/7.5 s grids: maximum difference 0.2375 K; response-node difference 0.04169 K.

Refinement changes sampled forcing; adaptive internal steps do not replace this check.

No temperature-error tolerance was declared, so no convergence pass is assigned. Undefined lag is not comparable.



All-node statistics

Temperature: K; capacity: J/K; configured power: W. Means are time-weighted.

1: Forward face

Min 225.028 | Mean 248.909 | Max 283.724 | Full range 58.696

Capacity 6200 | Internal power 0 | End minus start 5.360 K

2: Aft face

Min 225.225 | Mean 249.284 | Max 291.666 | Full range 66.441

Capacity 6200 | Internal power 0 | End minus start 4.068 K

3: Orbit-left face

Min 211.359 | Mean 212.005 | Max 212.689 | Full range 1.330

Capacity 7000 | Internal power 0 | End minus start 0.295 K

4: Orbit-right face

Min 231.574 | Mean 259.904 | Max 279.564 | Full range 47.990

Capacity 7000 | Internal power 0 | End minus start -9.101 K

5: Nadir face

Min 252.850 | Mean 259.558 | Max 265.379 | Full range 12.529

Capacity 8500 | Internal power 0 | End minus start -4.866 K

6: Zenith face

Min 229.912 | Mean 262.137 | Max 301.864 | Full range 71.952

Capacity 8000 | Internal power 0 | End minus start -33.415 K

All-node statistics (continued)

7: Spacecraft structure

Min 247.871 | Mean 251.652 | Max 255.642 | Full range 7.771

Capacity 26000 | Internal power 2 | End minus start -1.160 K

8: GNSS antenna

Min 245.796 | Mean 262.748 | Max 280.520 | Full range 34.724

Capacity 3200 | Internal power 0.4 | End minus start -13.522 K

9: GNSS RF front end

Min 258.937 | Mean 261.946 | Max 264.492 | Full range 5.556

Capacity 1800 | Internal power 5 | End minus start 0.131 K

10: GNSS oscillator

Min 261.160 | Mean 262.727 | Max 264.206 | Full range 3.046

Capacity 320 | Internal power 0.8 | End minus start 0.642 K

11: Receiver digital board

Min 263.121 | Mean 264.036 | Max 264.940 | Full range 1.819

Capacity 5200 | Internal power 12 | End minus start 0.582 K

Inputs and result details

Scenario name: reference_600km_24h_report

Duration: 86400 s

Time step: 60 s

Fixed warmup setting (non-convergence mode): 30 orbit

Node count: 11 node

Conductance range: [0, 1.4] W/K

Internal power range: [0, 12] W

Start epoch UTC: 2024-03-20 12:00:00

Altitude: 600000 m

Inclination: 97.6 deg

Attitude mode: nadir

Trajectory frame: Not available

Status: Not available

Acceptance and interpretation boundary

Numerical and configuration checks

Finite scenario temperatures: Pass

Telemetry validation: not performed.

Interpretation limits

The reference network is a research example, not flight qualification.

A completed computation is not physical validation; missing evidence is not a pass.

Report conclusion: numerical and input checks are above; physical validity needs independent evidence.

Audit files

MAT results, metric CSV, Markdown, manifest, and figures accompany this PDF.

Software version: 1.2.3